Workshop 5: Data analysis using AI/ML

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Time Table

- 13.00 – 13.30 Round 0 – *Introduction*

13.30 – 14.30 Round 1 – *RandomForest classifier*

Break 15 minutes

- 14.45 – 15.15 Round 2 – *RandomForest regressor*

15.15 – 16.00 Round 3 – *Both RF classifier & Regressor*

16.00 – 16.30 Round 4 – *Apply models from round 2 and 3*

Datasets:

Rounds

1. LIT-PCBA: ALDH1
2. OpenADMET Blind Challenge: Predicting CYP Inhibition
3. GPCRdb – 5ht2a_human ligands
4. Your designs from day 2

Sources:

[1] https://drugdesign.unistra.fr/LIT-PCBA/
[2] https://huggingface.co/spaces/openadmet/cyp-challenge
[3] https://gpcrdb.org/services/ligands/5ht2a_human/

Round 0

- Claude setup working – Claude code not Claude chat!
- Run the imports and the data loading in the round 1 seed
- You should see 107,237 molecules, 5,363 actives

Tasks

- Create the environment and install the requirements.
- Run the loading cells of the round 1 seed.
- Confirm you see 107,237 molecules, 5,363 actives or similar.
- Ask Claude to explain one thing in the file you don't recognise.

Round 1 – Classification

- Virtual screening is a needle-in-a-haystack problem
- Call everything inactive and you score 95% accuracy, and find nothing
- Enrichment factor: how many real hits land in the top 1% you could afford to buy
- So the metric, not the accuracy, decides whether the model is useful

Tasks

- Featurize the molecules: descriptors and ECFP4 fingerprints.
- Split the data into training-validating sets.
- Train a random forest on descriptors, on fingerprints, and on both.
- Evaluate the three models.
- Visualise the results.

EF1%: how much richer in hits the top 1% is than random picking

Worked example: 100,000 inactives and 5,000 actives, so 105,000 molecules and a 4.8% random hit rate

How EF1% is calculated

EF1% = hit rate in the top 1% ÷ hit rate in the whole library

1.0 means no better than random picking

1. Rank all 105,000 molecules by predicted score
2. Take the top 1%: 1,050 molecules
3. Count how many actives sit in that slice
4. Compare with the 50 you would get at random
5. 400 actives found means EF1% = 8

Three models on the same library

Model	Actives in top 1%	EF1%
Ranks no better than random	50	1.0
Useful ranker	400	8.0
Perfect ranker	1,050	21.0

Why accuracy misleads

Calling every molecule inactive scores 95.2% accuracy, yet its top 1% holds 50 actives, exactly what random picking gives. EF1% says 1.0.

LIT-PCBA: a benchmark built for machine learning

Tran-Nguyen, Jacquemard & Rognan, J. Chem. Inf. Model. 2020, 60, 4263–4273

15
protein targets,

7,844
confirmed actives from dose-response assays

407,381
confirmed inactives, measured not assumed

Measured, not invented

149 confirmatory dose-response PubChem bioassays. Inactives are experimentally tested compounds, not computational decoys.

Artifacts stripped out

Frequent hitters, autofluorescent compounds and others removed. Nearly 60% of raw actives were discarded.

Property matched

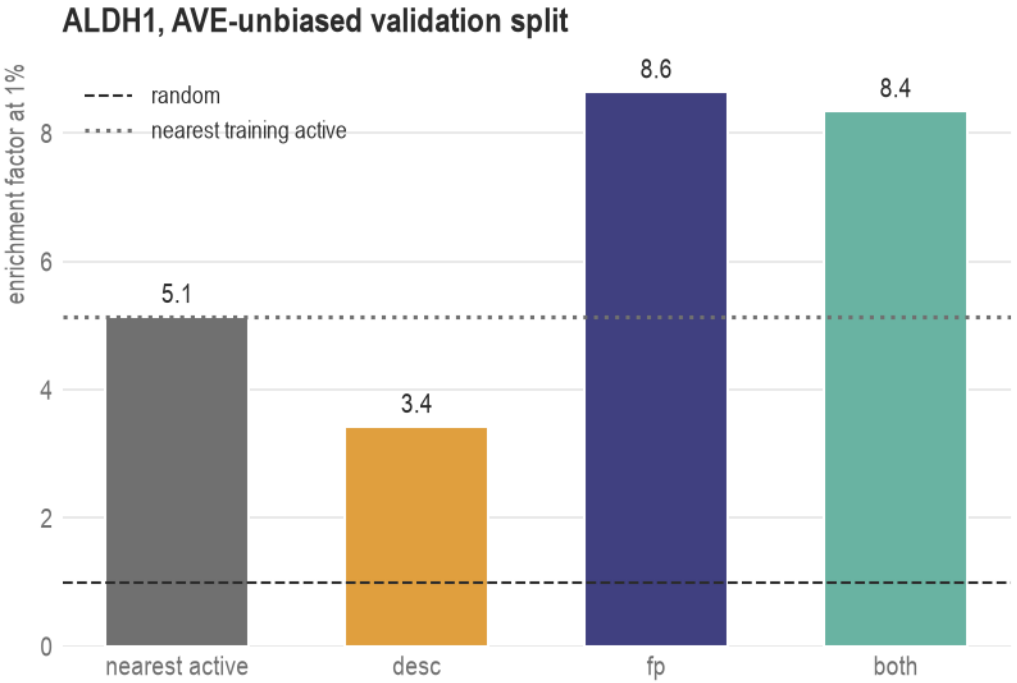
Actives and inactives span the same ranges of molecular weight, logP, charge, rotatable bonds and hydrogen bond counts.

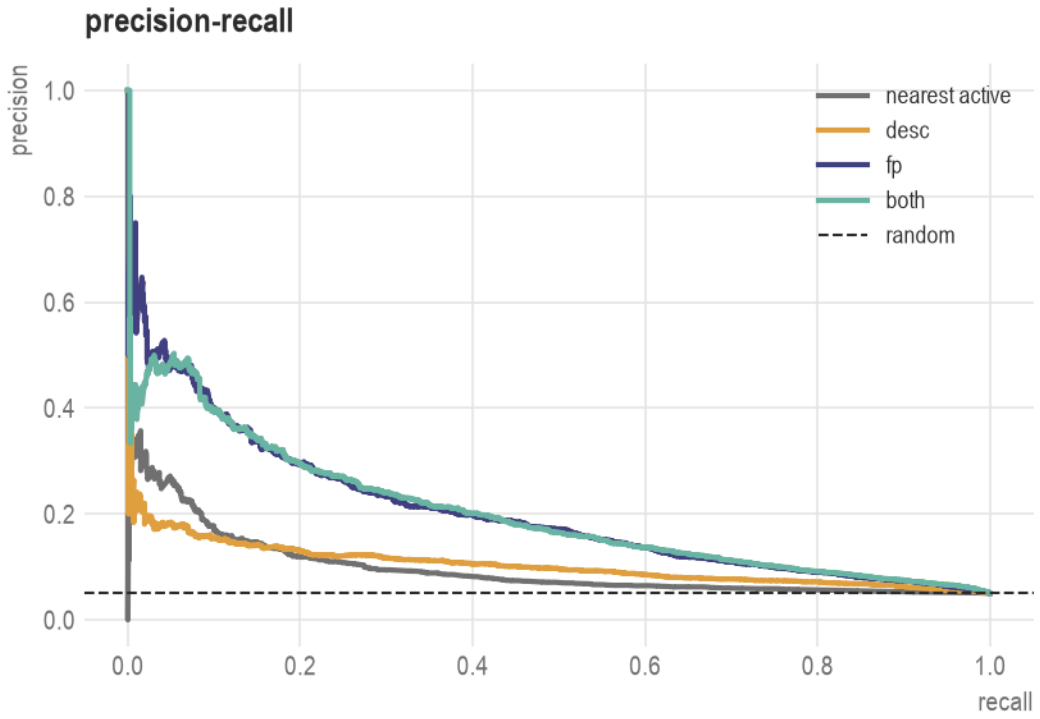
Splits already debiased

Training and validation were unbiased with AVE, asymmetric validation embedding, which rebalances nearest-neighbour distances. To avoid memorisation.

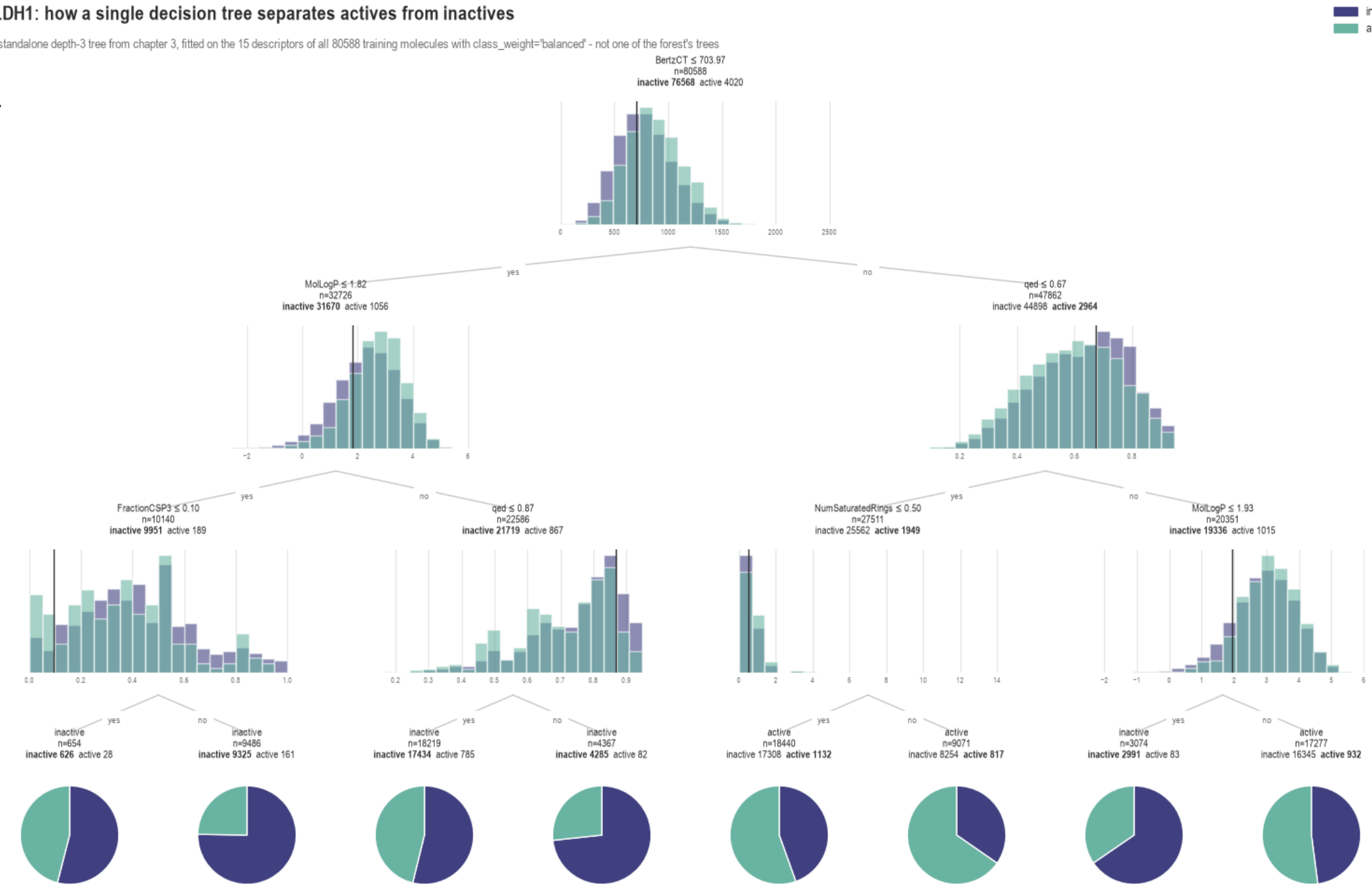
Your task Use **ALDH1**, the largest target in the set and one of seven the authors recommend as having enough actives to train on. Build a random forest classifier only ligand based.

Round 1 - Results





Round 1


ALDH1: how a single decision tree separates actives from inactives
the standalone depth-3 tree from chapter 3, fitted on the 15 descriptors of all 80588 training molecules with class_weight="balanced" - not one of the forest's trees
inactive
active
BertzCT ≤ 703.97
n=80588
inactive 76568 active 4020
yes
no
MolLogP ≤ 1.82
n=32726
inactive 31670 active 1056
qed ≤ 0.67
n=47862
inactive 44898 active 2964
FractionCSP3 ≤ 0.10
n=10140
inactive 9951 active 189
qed ≤ 0.87
n=22586
inactive 21719 active 867
NumSaturatedRings ≤ 0.50
n=27511
inactive 25562 active 1949
MolLogP ≤ 1.93
n=20351
inactive 19336 active 1015
inactive
n=654
inactive 626 active 28
inactive
n=9486
inactive 9325 active 161
inactive
n=18219
inactive 17434 active 785
inactive
n=4367
inactive 4285 active 82
active
n=18440
inactive 17308 active 1132
active
n=9071
inactive 8254 active 817
inactive
n=3074
inactive 2991 active 83
active
n=17277
inactive 16345 active 932

Five years on, the benchmark was audited

Huang, Knight & Naprienko, arXiv:2507.21404, 2025. An audit of data leakage in LIT-PCBA

They found out the dataset is not as unbiased as claimed!

2,491

inactives appear in both the training and the validation split, across the 15 targets

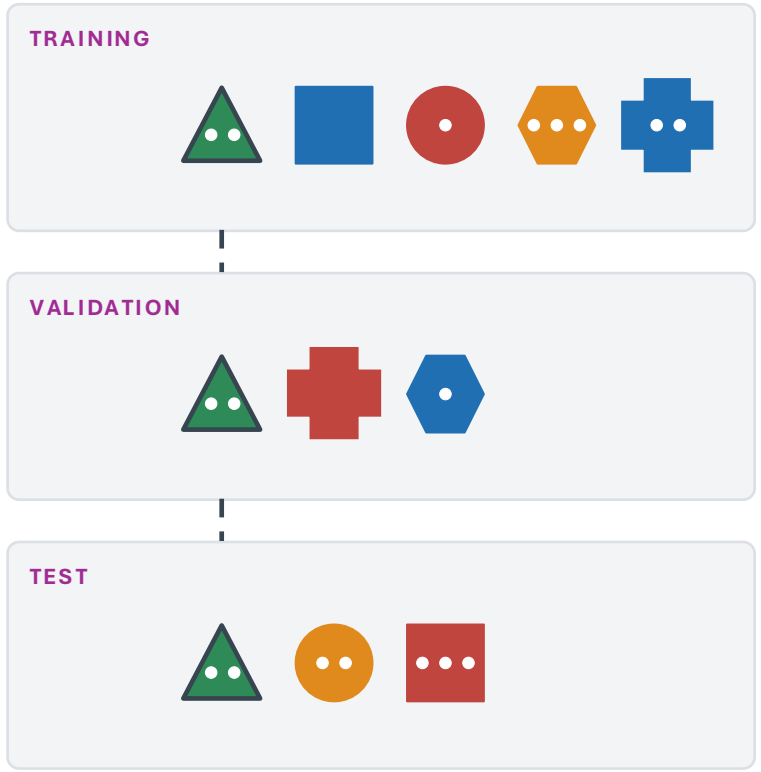
323

ALDH1 active pairs spanning training and validation with ECFP4 similarity above 0.6

What leakage looks like

1 Random split

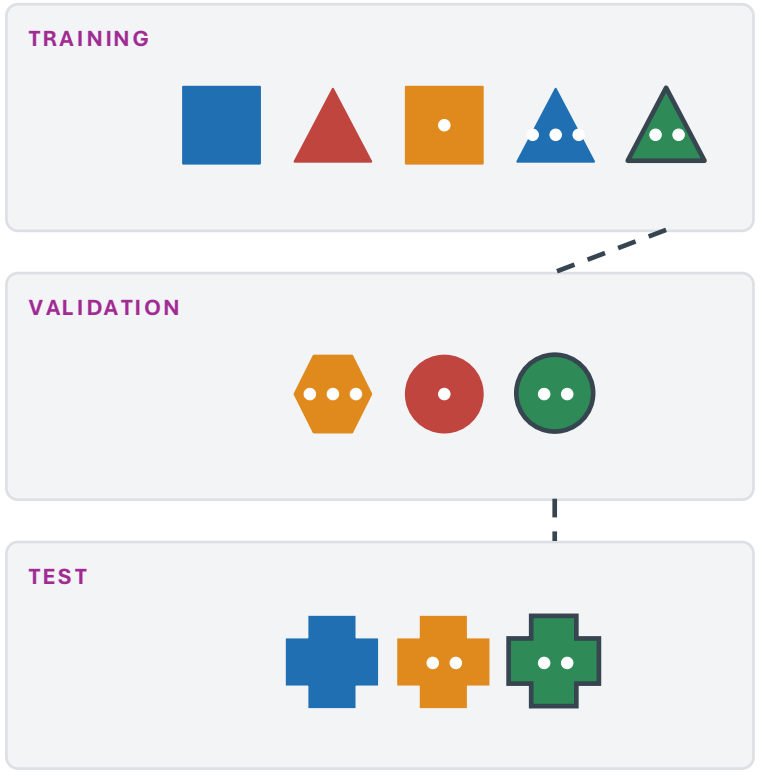
Deduplicated. No repeats anywhere.



Three different molecules sharing every feature a fingerprint sees. Deduplication cannot touch them.

2 Split by scaffold

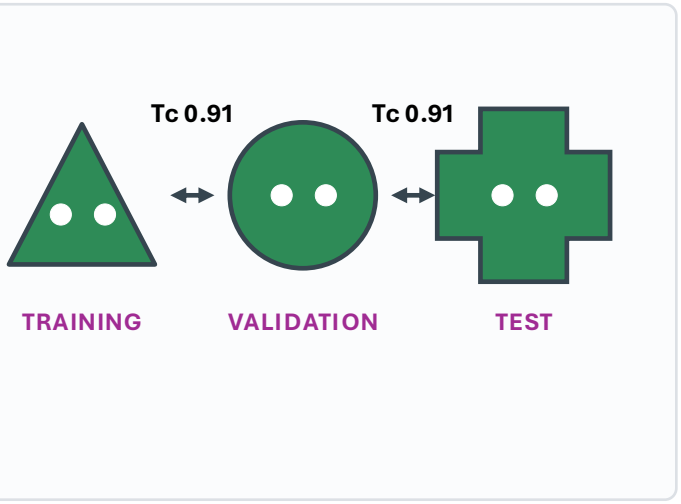
Each shape family sits in one set only.



No shape appears twice. The rule looks satisfied, yet the same family still reaches both held-out sets.

3 What the rule missed

Scaffold differs. Everything else carries over.



The filter separated the scaffold.
Colour and dots crossed into validation and test untouched, so both held-out sets still contain near neighbours of the training data.

In ALDH1, 323 such pairs survived the benchmark's own debiasing step.

shape scaffold family

colour a second property

dots a third property

four property values

The split is the hard part

What to carry into your own virtual screening work

01	**Careful is not the same as clean**	LIT-PCBA's authors applied the best debiasing method available, asymmetric validation embedding (AVE), and reported their bias values honestly. The leakage still survived, because AVE minimises an aggregate statistic over distance distributions and never forbids any individual pair from being near-identical.
02	**Some benchmarks cannot be repaired**	The audit's authors set out to publish a corrected LIT-PCBA and decided against it. Removing the analog leakage and the low-diversity query sets would mean discarding much of the data and redefining the splits, which amounts to building a new benchmark rather than fixing this one.
03	**Check the thing you actually care about**	A scaffold rule constrains scaffolds. If what matters is that no held-out molecule has a close neighbour in training, threshold that similarity directly and verify it afterwards. Run a memorization baseline: if nearest-neighbour lookup matches your model, the split is doing the work.
04	**Read the failures before repeating them**	Both papers are short. The 2020 paper shows what careful dataset design looks like; the 2025 audit shows what it still missed. Between them they cover most of the ways a screening benchmark goes wrong.

Tran-Nguyen, Jacquemard & Rognan, J. Chem. Inf. Model. 2020, 60, 4263–4273
Huang, Knight & Naprienko, arXiv:2507.21404, 2025 · code at github.com/sievestack/LIT-PCBA-audit

Round 2 – Regression

Predicting CYP inhibition on the OpenADMET challenge's own train/test split

- Predict the pIC50 for four CYP isoforms: 1A2, 2C9, 2D6, 3A4
- Inhibit a CYP and you cause drug-drug interactions
- So predict the liability before you ever make the compound
- A deliberately simple regressor on the challenge's own train/test split
- Compare against the leaderboard, and save the model for round 4

Tasks

- Featurize both splits.
- Inspect the data.
- Train one regressor per isoform. Which CYP is hardest, and why?
- Use regression metrics against a mean baseline: can you beat it?
- Save the cyp.joblib for round 4.

Dataset: OpenADMET: Predicting CYP Inhibition

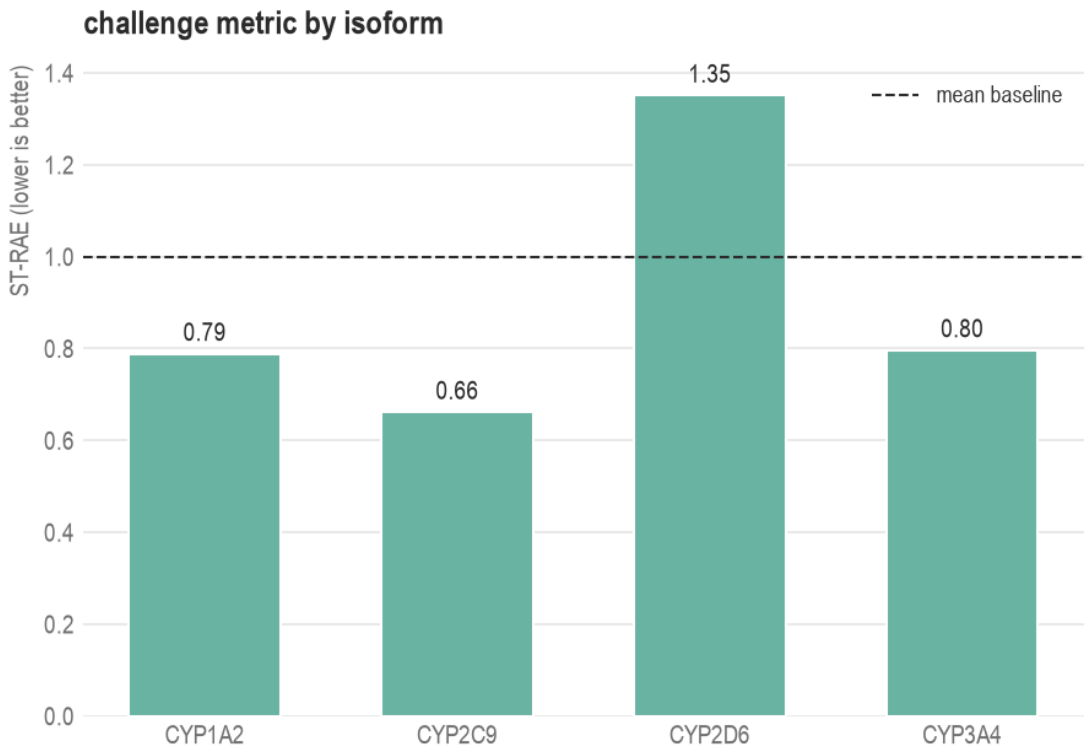
What it measures:
Inhibition of four drug-metabolising enzymes: CYP1A2, CYP2C9, CYP2D6 and CYP3A4. Label is pIC50, the negative log of the half-inhibiting concentration, so higher means a stronger inhibitor and one unit is tenfold.

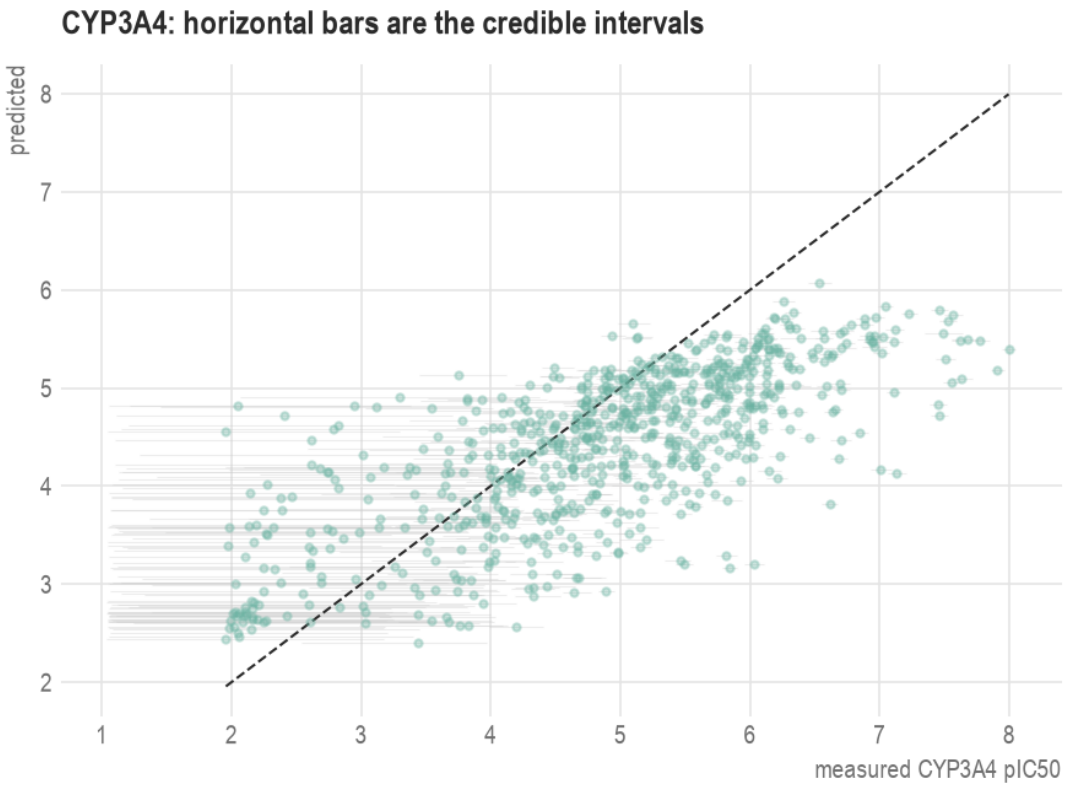
Size and split
4,905 training molecules and 750 test molecules, split as the organisers built it. Columns are SMILES, one …_pIC50_direct_inhibition per enzyme, plus a credible interval for each value.

Where it comes from
A screen of commercial libraries, with the hits promoted to full 12-point dose-response curves. The test set is close chemical analogs of the strongest hits, so it is dense in activity cliffs and not a random sample.

Two catches
Labels are missing for many molecule and enzyme combinations, so each target needs its own mask. Values below pIC50 4 fall under the lowest tested dose and are noisy, which is why the credible intervals are supplied

Round 2 - Results





Round 3 – Case study: 5-HT2A

Measured ligands of the human 5-HT2A receptor, pulled from GPCRdb

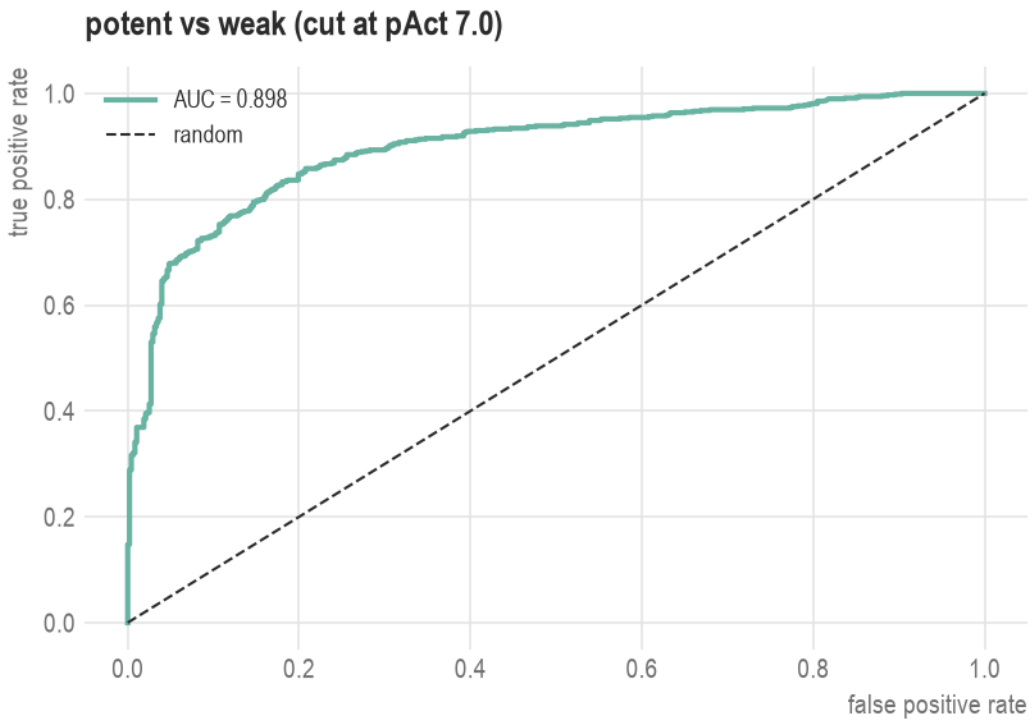
- Measured 5-HT2A ligands, pulled straight from GPCRdb
- No inactives in the set: the weakest thing in there still binds
- So split the known ligands at a potency threshold instead
- Classifier: potent or weak binder?
- Regressor: binding affinity, pKi
- Save both models for round 4

Tasks

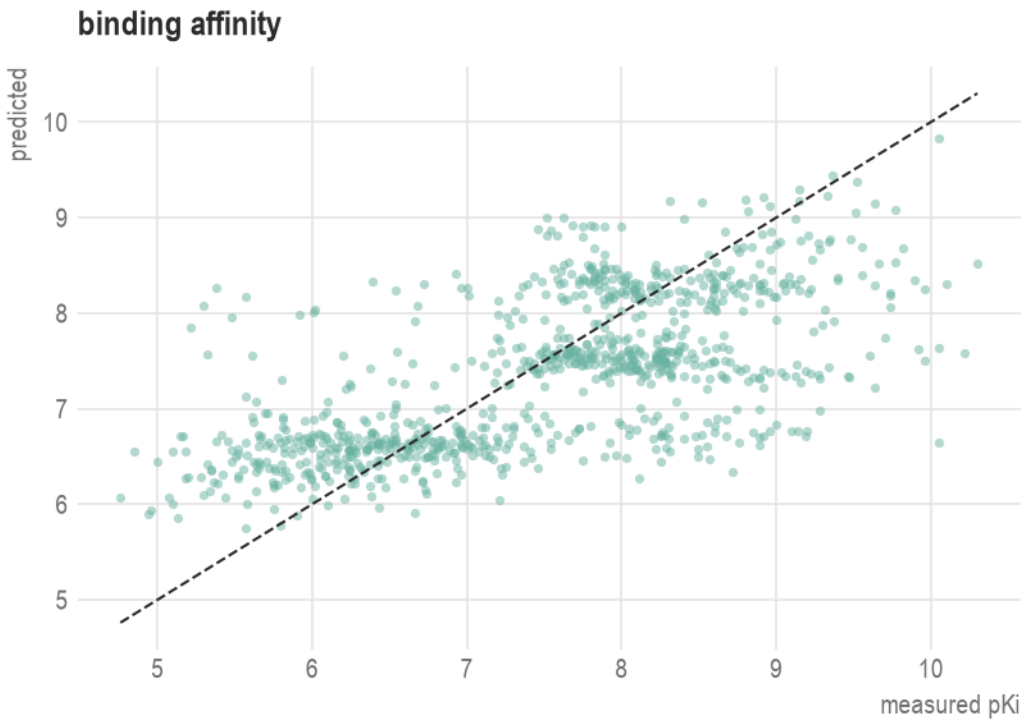
- Inspect the data.
- Train the classifier at a potency threshold of your choosing.
- Train the pKi regressor.
- Evaluate the results of the models.
- Save 5ht2a_classifier.joblib, 5ht2a_regressor.joblib and the reference ligands for round 4.

Round 3 - Results

Random Forest Classifier



Random Forest Regressor



Round 4 – Score your own designs

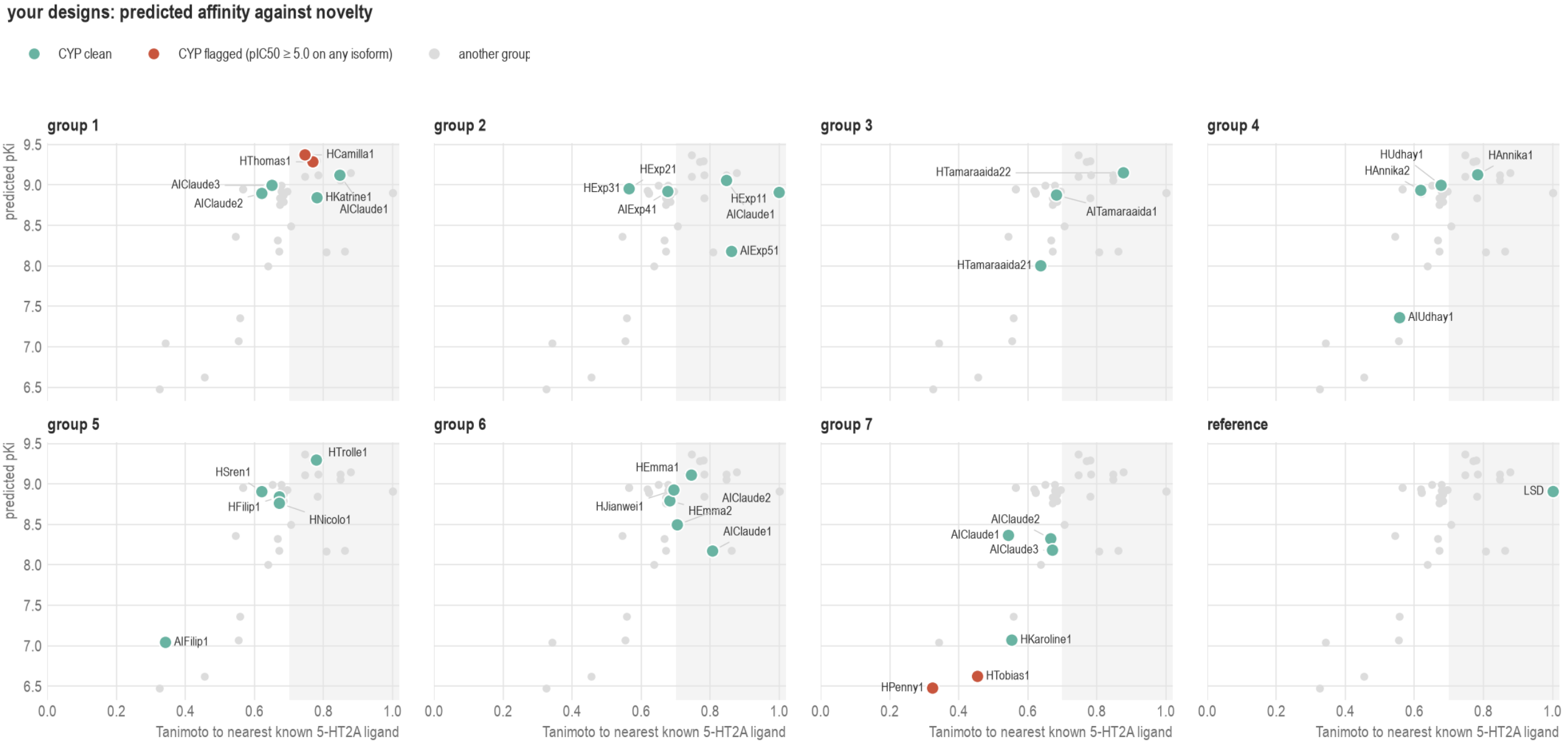
Your day 2 molecules, run through the models you trained in rounds 2 and 3

- Your designs: your_designs/workshop_designs.smi
- The seed loads your models from rounds 2 and 3
- Potent or weak at 5-HT2A, and predicted pKi
- CYP inhibition across all four isoforms
- Novelty against the 5-HT2A ligand set (Tanimoto similarity)

Tasks

- Featurize the new compounds.
- Compute novelty of your designs.
- Make predictions from your models.

Round 4 - Results



Round 4 – Predictions for your designs

36 designs ranked by predicted pKi at 5-HT2A, with the classifier's potency call and its probability

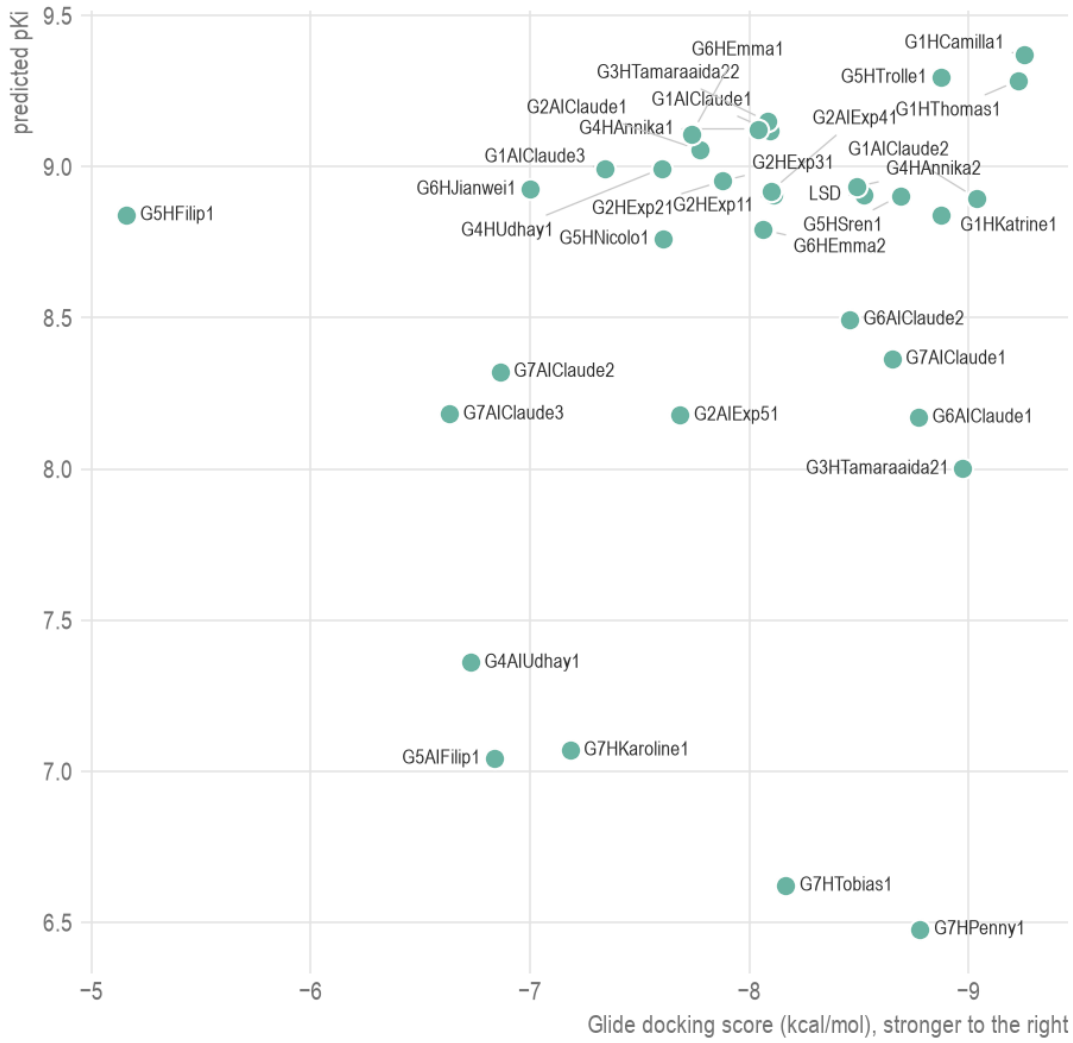
Name	Call	P potent	pKi pred
G1HCamilla1	potent	0.68	9.37
G5HTrolle1	potent	0.79	9.29
G1HThomas1	potent	0.79	9.29
G3HTamaraaida22	potent	0.93	9.15
G4HAnnika1	potent	0.88	9.13
G1AIClaude1	potent	0.88	9.12
G6HEmma1	potent	0.83	9.11
G2AIClaude1	potent	0.90	9.06
G4HUdhay1	potent	0.72	9.00
G1AIClaude3	potent	0.77	8.99
G2HExp21	potent	0.79	8.95
G2HExp31	potent	0.79	8.95

Name	Call	P potent	pKi pred
G4HAnnika2	potent	0.74	8.93
G6HJianwei1	potent	0.79	8.93
G2AIExp41	potent	0.81	8.92
G2HExp11	potent	0.95	8.91
LSD	potent	0.95	8.91
G5HSren1	potent	0.82	8.90
G1AIClaude2	potent	0.83	8.90
G3AITamaraaida1	potent	0.82	8.87
G1HKatrine1	potent	0.77	8.84
G5HFilip1	potent	0.82	8.84
G6HEmma2	potent	0.81	8.79
G5HNicolo1	potent	0.76	8.76

Name	Call	P potent	pKi pred
G6AIClaude2	potent	0.71	8.49
G7AIClaude1	potent	0.64	8.36
G7AIClaude2	potent	0.74	8.32
G7AIClaude3	potent	0.61	8.18
G2AIExp51	potent	0.84	8.18
G6AIClaude1	potent	0.81	8.17
G3HTamaraaida21	potent	0.78	8.00
G4AIUdhay1	potent	0.66	7.36
G7HKaroline1	potent	0.58	7.07
G5AIFilip1	potent	0.56	7.04
G7HTobias1	weak	0.40	6.62
G7HPenny1	weak	0.33	6.48

Round 4 – Comparison docking results

5HT2A



5HT2B

